VisionGuard: Smart Cane for Visually Impaired People

PROJECT REPORT

School of Computer Science and Engineering (SCOPE) – B.Tech CSE

Vellore Institute of Technology, Vellore

BACSE324 | Embedded Systems Architecture

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Submitted by

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1. Team Members

Enter the full names and registration numbers of all team members.

S. No.	Name	Registration Number
1	S Darshan	25BCE2883
2	Tanay Agarwal	25BCE2475

2. Objectives

State the main goals of the project clearly and briefly. Use measurable objectives where possible.

- Detect obstacles in front of the cane within **50 cm** using an HC-SR04 ultrasonic sensor (range 2–400 cm), so the user is warned before hitting them.
- Show how close the obstacle is through the beep speed: the gap between beeps goes from **50 ms at 2 cm** to **400 ms at 50 cm**.
- Detect water or wet ground at the tip when the moisture reading drops below a set value (**500 out of 1023**), and give a continuous tone that has **priority** over the obstacle beeps.
- Keep the cane cheap and portable: it runs on one **18650 battery** with an on/off switch, and the cane parts cost **under ₹900**.

3. Working Principle

Briefly explain how the project works. Describe the relevant physics principles, the input–processing–output sequence, and how the hardware/software components work together.

Input → processing → output. The cane has two inputs: the HC-SR04 ultrasonic sensor, which measures the distance in front, and a moisture probe with an LM393 module, which senses wetness at the tip. The Arduino Nano does the processing. It times the echo, works out the distance, reads the moisture voltage with its 10-bit ADC and decides which alert to give. The output is an active buzzer on pin D2. Everything is powered by one 18650 battery, connected through a rocker switch to the Nano's 5V pin, which also powers both sensors.

3.1 Physics Involved

(a) Ultrasonic distance measurement. The Nano sets the Trig pin HIGH for 10 µs. The HC-SR04 then sends out eight 40 kHz sound pulses and keeps the Echo pin HIGH until the echo comes back. The time Echo stays HIGH (t) is the time for the sound to reach the object and come back, so the distance is:

$$d = (v \times t) / 2$$

Here v is the speed of sound, about 343 m/s at 20 °C, which is 0.0343 cm/µs (our code uses 0.034). For example, if t = 1000 µs, then d = 0.034 × 1000 / 2 = 17 cm. The speed of sound changes slightly with temperature ($v \approx 331.3 + 0.606 \cdot T$ m/s), which gives an error of about 0.2% per °C. The sensor's beam is a cone of about 15°, so it works best on hard, flat surfaces facing it. Soft or angled surfaces can absorb the sound or bounce it away. We set a 30 ms timeout in pulseIn(), which covers about 5 m there and back.

(b) Moisture sensing. The probe has two bare copper tracks. Dry air between them does not conduct, so the resistance R_p is very high. Water has dissolved salts and conducts, so R_p drops a lot. On the module, the probe and a fixed resistor R_f form a voltage divider across V_cc, and the A0 pin gives the middle voltage:

$$V_{A0} = V_{cc} \times R_p / (R_p + R_f) \qquad N = 1023 \times V_{A0} / V_{ref}$$

N is the 10-bit ADC value. When dry, R_p is very large, so V_A0 is close to V_cc and N is about 1023. When wet, R_p is small, so V_A0 and N go down. Since the ADC reference and the divider use the same supply, the reading depends on the resistance ratio and not on the exact battery voltage (it is ratiometric). We do not use the LM393's digital D0 output; we set the threshold in the code instead.

(c) Buzzer. We use an active buzzer, which has its own oscillator and a piezo disc inside. When the GPIO pin goes HIGH, the disc vibrates at a few kHz and makes a tone.

(d) Power. The 18650 Li-ion cell is 4.2 V when full and about 3.0 V when empty (3.7 V nominal). The ATmega328P and our version of the HC-SR04 both work in this range. The capacitor across the supply stores charge and gives the extra current needed when the buzzer switches on ($i = C \cdot dV/dt$), so the voltage does not dip much. The cane draws roughly 30–40 mA, so a 1200 mAh cell should last about 30 hours in theory (1200 mAh ÷ 40 mA).

3.2 Algorithm / Program Logic

1. **Start-up:** set D5 (Trig), D2 (buzzer) and D3 (kept for the vibration motor) as outputs and D6 (Echo) as an input. Start Serial at 9600 baud and switch the buzzer off.
2. **Measure distance:** make Trig LOW for 2 µs, HIGH for 10 µs, then LOW again. Measure how long Echo stays HIGH using pulseIn() with a 30 ms timeout (it returns 0 if there is no echo). Then d = t × 0.034 / 2 cm.
3. **Read moisture:** m = analogRead(A0), which gives a value from 0 to 1023.
4. **Print:** show d, m and the current state on the Serial Monitor.
5. **Check water first:** if m < 500, go to the WATER state and keep the buzzer ON.
6. If not, and 0 < d < 50 cm, go to the OBSTACLE state. Limit d to 2–50 cm, find the gap g = map(d, 2, 50, 50, 400) ms, beep for 50 ms and then stay quiet for g ms. A closer obstacle gives a shorter gap, so the beeps get faster.
7. Otherwise (d = 0, meaning no echo, or d ≥ 50 cm), go to the CLEAR state and turn the buzzer off.
8. **Repeat:** wait 30 ms so the old echo dies out, then go back to step 2. This keeps running as long as the cane is on.

What should happen: in open space the cane stays quiet. When a hand moves in from 60 cm, the beeping starts at 50 cm and gets faster as the hand comes closer. When the tip is dipped in water, the tone becomes continuous, even if there is also an obstacle in front.

3.3 Data Flow Graph (DFG)

Insert a DFG showing the computational steps and dependencies between operations. Label inputs, operations, intermediate variables, and outputs.

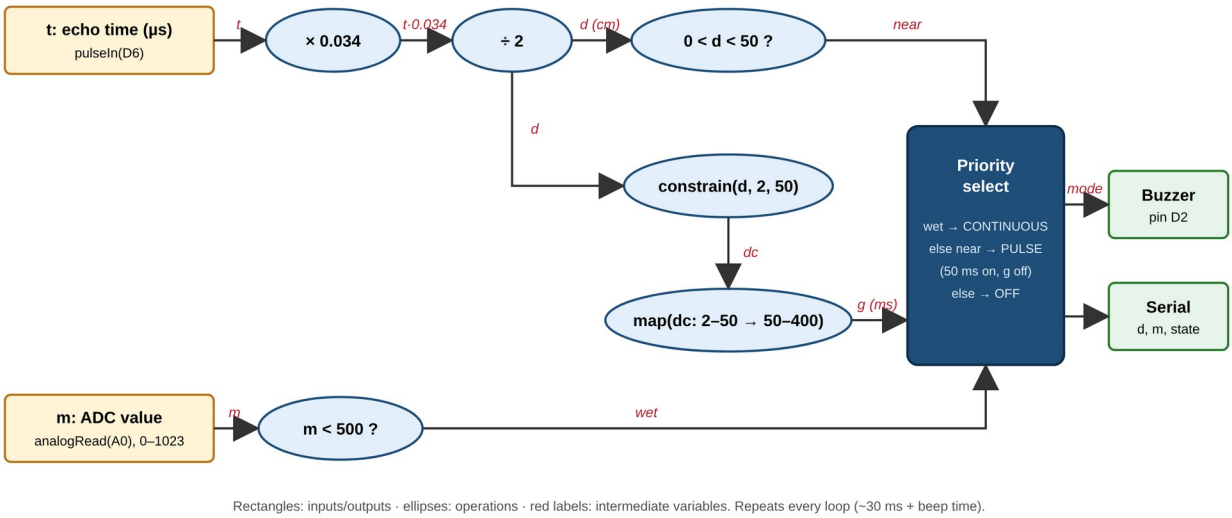


Figure 1. Data Flow Graph of the proposed system.

The echo time t goes through two steps (× 0.034 and ÷ 2) to give the distance d. The distance is then used in two places: a check for whether the obstacle is near, and the constrain and map steps that give the beep gap g. The moisture value m is compared with 500 to decide whether it is wet. The priority block takes wet, near and g and controls the buzzer. The two sensor paths do not depend on each other, so they can be worked out in any order.

3.4 State Diagram

Insert a state diagram showing the system states, transition conditions, and actions performed in each state.

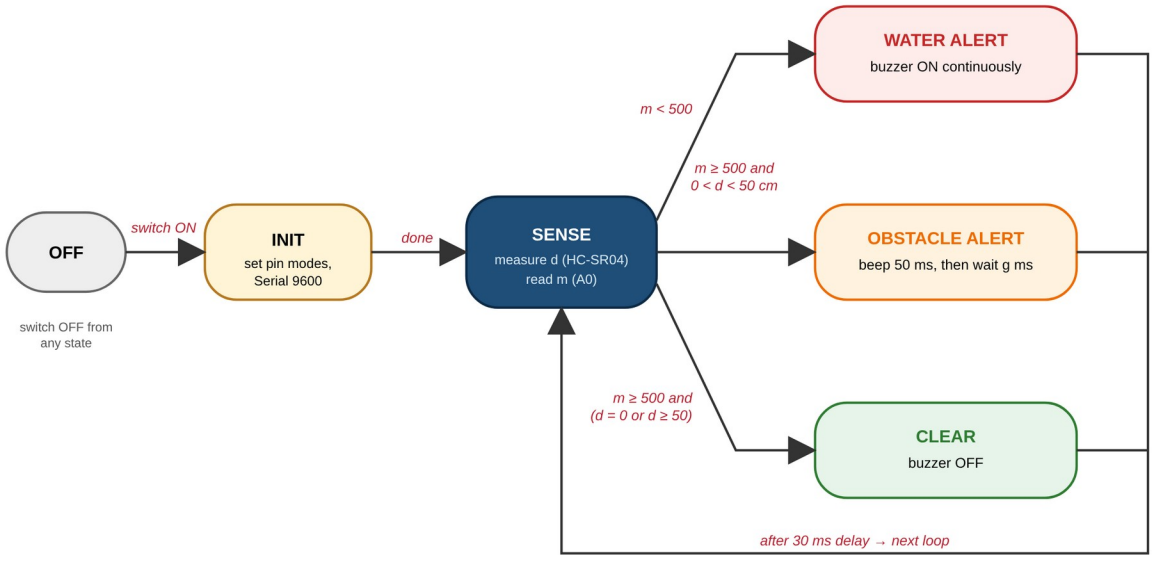


Figure 2. State diagram of the system.

When the cane is switched on, it sets up once and then goes to SENSE in every loop. From SENSE it goes to exactly one of three states: WATER (m < 500), OBSTACLE (m ≥ 500 and 0 < d < 50 cm) or

CLEAR (everything else). Each of these goes back to SENSE after the 30 ms delay. Turning the switch off takes it back to OFF from any state.

4. Final Circuit Diagram

Insert a clear, labelled circuit diagram of the final implementation. Show component names, pin connections, supply voltages, ground connections, and relevant component values.

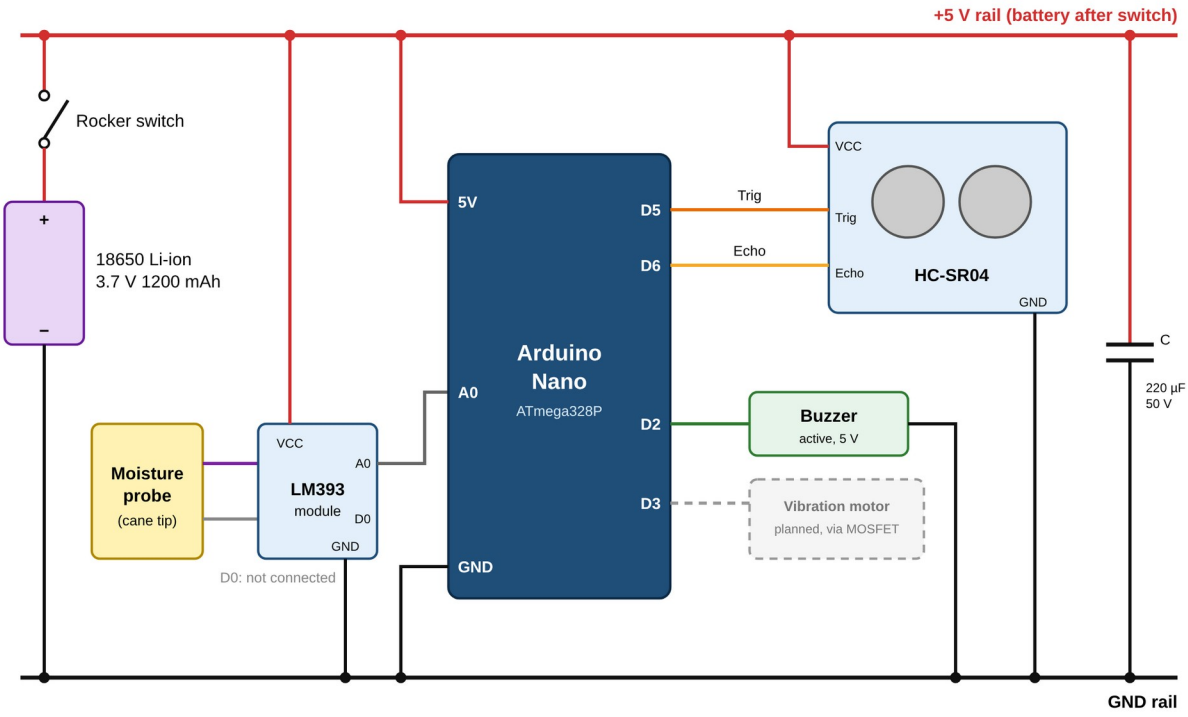


Figure 3. Final circuit diagram.

Table 1. Pin connections.

Device pin	Arduino Nano pin	Notes
18650 cell + (via rocker switch)	5V	3.0–4.2 V (3.7 V nominal) on the 5V rail
18650 cell −	GND	Common ground
HC-SR04 VCC / GND	5V / GND	
HC-SR04 Trig	D5	10 µs trigger pulse (output)
HC-SR04 Echo	D6	Echo pulse width (input)
LM393 module VCC / GND	5V / GND	
LM393 module A0	A0	Analog moisture value, 0–1023
LM393 module D0	Not connected	Threshold applied in software
Moisture probe (2 wires)	LM393 probe header	Non-polarised
Active buzzer + / −	D2 / GND	HIGH = ON
Electrolytic capacitor + / −	5V / GND	220 µF, 50 V supply filter
Vibration motor (planned)	D3 via IRLZ44N MOSFET	Not fitted in this prototype

5. Photograph of the Final Product

Insert a clear photograph of the completed prototype. Prefer a well-lit image with the important components visible. Add labels or multiple views if they help explain the build.



Figure 4. Photograph of the completed project. Top: the complete cane, with the moisture probe at the top end, then the LM393 module, HC-SR04 sensor, Arduino Nano with buzzer and 220 µF capacitor, rocker switch and 18650 battery. Bottom left: moisture probe at the tip. Bottom right: rocker switch and battery holder.

6. Conclusion

Summarise the project outcome using the following prompts:

- Objectives achieved: On battery power the cane beeps when an obstacle comes within 50 cm, and the beeps get faster as it comes closer (objectives 1 and 2). We checked this by moving a hand or a notebook towards the sensor while watching the distance on the Serial Monitor. The water alert (objective 3) works in the code and has priority over obstacles; dry reading ≈ dry value, wet reading ≈ wet value, threshold set to value. The cane runs on one 18650 cell with a switch, and the cane parts cost ₹876 (objective 4).
- Challenges faced: The 3.7 V cell is connected straight to the 5V pin without a regulator, so the readings can become unreliable as the battery runs down. The ultrasonic beam misses thin, soft and head-level objects. In the first version of the code, a missed echo stopped the loop for up to 1 second because of the default pulseIn() timeout; we fixed this with a 30 ms timeout. The moisture threshold also has to be set separately for each probe.
- Improvements: Add a 5 V boost converter, a TP4056 charging module and the 1 A fuse for steady, safe power. Average a few distance readings and correct for temperature to improve accuracy. Use a capacitive moisture sensor so the probe does not corrode. Add a vibration motor (through the IRLZ44N MOSFET) so alerts can be felt in noisy places. Move the circuit onto a PCB inside a waterproof box.
- Future scope: Add more ultrasonic sensors pointing up (for head-level objects) and down (for steps and potholes), GPS with GSM or Bluetooth to send an SOS message to family, a low-battery warning, and trials with visually impaired users to find the best alert distance (for example 100–150 cm).

VisionGuard shows that a plain cane can be turned into a cheap electronic travel aid using an Arduino Nano, an ultrasonic sensor and a moisture probe. It warns about obstacles before the user touches them, shows the distance through the beep speed, and gives a priority warning for water, all from one rechargeable battery. The prototype meets its main objectives in our bench tests. The main things to improve are regulated and protected power, vibration feedback and wider sensing, which would make it ready for real-world testing.

7. References

List every source used, including textbooks, research papers, datasheets, official documentation, websites, and software libraries. Use one consistent citation style (IEEE or APA).

[1] Elecfreaks, “Ultrasonic Ranging Module HC-SR04,” datasheet, 2011. [Online]. Available: https://cdn.sparkfun.com/datasheets/Sensors/Proximity/HCSR04.pdf

[2] Microchip Technology, “ATmega48A/PA/88A/PA/168A/PA/328/P megaAVR Data Sheet,” DS40002061, 2020.

[3] Arduino, “Arduino Nano,” Arduino Documentation. [Online]. Available: https://docs.arduino.cc/hardware/nano

[4] Arduino, “Language Reference: pulseIn(), analogRead(), map(), constrain(),” Arduino Documentation. [Online]. Available: https://www.arduino.cc/reference/en/

[5] Texas Instruments, “LMx93-N Low-Power, Low-Offset Voltage, Dual Comparators,” datasheet. [Online]. Available: https://www.ti.com/product/LM393

[6] World Health Organization, “World Report on Vision,” Geneva, 2019.